



Faculty of Engineering and Natural Sciences
Computer Engineering Department

HEURISTIC OPTIMIZATION ALGORITHMS

Term Project Report — Spring 2026

Multi-Objective Diet Optimization Problem

An Evolutionary Multi-Objective Optimization Approach

Student 1	Saed O. S. Radi 2321051365
Student 2	Kenan Alemam 2521051363
Student 3	Ahmed Alsaleh 2321051353
Student 4	Abdulrahman Aljabahji 2221251353
Student 5	Mamdouh Al Masri 2221251392
Course	BLM20364E / BLM22332E — Heuristic Optimization Algorithms
Dataset	405 food items (MySQL database, 18 food groups)
Algorithms	NSGA-II and SPEA2 (pymoo library)
Objectives	User Preference (MAX), Cost (MIN), Preparation Time (MIN)
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1. Introduction

This report documents the design, implementation, and experimental evaluation of a Multi-Objective Diet Optimization Problem (MODP) solver. The goal is to recommend a personalised daily menu — composed of a breakfast and a lunch/dinner — selected from a curated MySQL database of 405 prepared food items.

The problem is formulated as a Multi-Objective Multidimensional Knapsack Problem (MOMKP) with three competing objectives: maximising user food preference, minimising meal cost, and minimising total preparation time. Since these objectives conflict with each other — high-preference foods are often expensive or slow to prepare — no single optimal solution exists. Instead, a Pareto front of non-dominated solutions is produced, offering a range of trade-off options.

Nutritional hard constraints derived from Dietary Reference Intakes (DRI) ensure that every recommended menu meets the target user's daily health requirements across five nutrients. Two evolutionary multi-objective algorithms are implemented and compared: NSGA-II and SPEA2, both using a fully custom chromosome representation, greedy decoder, and penalty function developed specifically for this problem.

2. Problem Definition

2.1 Mathematical Model

The MODP is modelled as a binary optimisation problem over $n = 405$ food items. Each variable $x_i \in \{0, 1\}$ indicates whether food item i is included in the menu.

Objective	Direction	Formula
User Preference	MAX (mandatory)	$f1(x) = \sum (x_i * \text{preference}_i)$
Cost	MIN	$f2(x) = \sum (x_i * \text{cost}_i)$
Preparation Time	MIN	$f3(x) = \sum (x_i * (\text{prepTime}_i + \text{cookTime}_i))$

Subject to: $RLL_j \leq \sum (x_i * \text{nutrient}(j,i)) \leq RUL_j$ for $j = 1..5$, $x_i \in \{0,1\}$

2.2 Nutritional Constraints (DRI)

Five nutrients must satisfy per-user Dietary Reference Intake bounds. A soft-constraint tolerance is applied to maintain a non-empty feasible space:

Nutrient	Unit	Soft Bounds Applied
Energy (C1)	kcal	Eff. Lower = $RLL \times 0.90$ Eff. Upper = $RUL \times 1.15$
Protein (C2)	g	Eff. Lower = $RLL \times 0.90$ Eff. Upper = $RUL \times 1.15$
Carbohydrate (C3)	g	Eff. Lower = $RLL \times 0.90$ Eff. Upper = $RUL \times 1.15$
Fiber total dietary (C4)	g	Eff. Lower = $RLL \times 0.90$ Eff. Upper = $RUL \times 1.15$
Sodium / Na (C5)	mg	Eff. Lower = $RLL \times 0.90$ Eff. Upper = $RUL \times 1.15$

Breakfast sub-constraints: $RLL_b = RLL \times 0.35$ | $RUL_b = RUL \times 0.35$ — applied only to Energy and Protein.

2.3 User Profiles

Two users from the database are evaluated independently throughout all experiments:

- User 1 (non-vegetarian) — has unrestricted access to all 405 food items from all food groups.
- User 2 (vegetarian) — automatically detected by analysing each user's preference ratings for meat-based food groups. A user is classified as vegetarian if more than 50% of their ratings for meat-based groups (Chicken Products, Meat Products, Chicken and Turkey Dishes, Fish, Meat-based Dishes) are -1. This reduces the candidate pool to approximately 314 items.
- Items carrying a per-user preference rating of -1 are unconditionally excluded via a penalty of 10^6 in objective evaluation — they must never appear in any menu.

2.4 Database Tables Used

All values are read directly from the MySQL database. No values are hardcoded:

Table	Key Columns Used
foods	id, name, foodGroupId, cost, preference, preparingTime, cookingTime, co2
user_foods	userId, foodId, preference — per-user ratings override foods.preference
food_nutrients	foodId, nutrientId, quantity — nutritional content per food item
nutrients	id, name, unit — nutrient names for the 5 DRI constraints
dri	userId, nutrientId, RLL, RUL — per-user daily intake bounds
food_group	id, name — 18 food groups used for diversity calculation

3. Chromosome Representation and Decoding

3.1 Two-Part Chromosome Design

Each candidate solution is encoded as two independent permutation chromosomes whose genes are food item IDs. The two parts are operated on independently — crossover and mutation are applied to each part separately:

Part	Food Groups Included
Breakfast (91 items)	Dairy Products, Jams and Syrups, Honey Products, Pancakes, Seed/Bean/Olive, Bakery, Sweet Things and Marmelades, Beverages, Cereals, Fruits, Beverages 2, Bakery 2
Lunch/Dinner (314 items)	Soups, Salads, Cooked, Chicken and Turkey, Vegetable Based, Fish, Meat-based Dishes, Pasta, Desserts, Rice and Bulgur, Nut and Seed, Appetizers, Soups 2, Salads 2, Appetizer 2

This design ensures that breakfast menus contain only culturally and nutritionally appropriate breakfast foods (~91 items), while lunch/dinner draws from a broader pool. The chromosome is a permutation of food IDs within each part.

3.2 Greedy Decoding (Genotype to Phenotype)

The chromosome encodes a priority ordering of food items — it is NOT the menu directly. A greedy left-to-right scan converts it to an actual selected menu:

Pass	Algorithm
Breakfast	Scan breakfast genes left-to-right. Tentatively add each item. Skip if adding it would push Energy OR Protein above 35% of daily RUL x 1.15. Stop when both Energy AND Protein reach 35% of daily RLL x 0.90.
Lunch / Dinner	Carry over nutrient totals from breakfast. Scan lunch/dinner genes left-to-right. Check all 5 nutrients cumulatively. Skip if any nutrient would exceed RUL x 1.15. Stop when ALL 5 nutrients meet their daily lower bound.
Excluded items	Any food item with preference = -1 is unconditionally skipped during decoding.

3.3 Diversity Mechanism (Option B — Penalty Term)

To prevent menus dominated by a single food group, a diversity penalty is added to the fitness evaluation. This is Option B from the project specification:

```
R_diversity = alpha / |distinct foodGroupId values in menu| (alpha = 1.0)
```

Target: at least 4-6 distinct food groups per daily menu. Experiment 3 compares results with (alpha = 1.0) vs without (alpha = 0.0) the diversity mechanism. The approach penalises solutions with low group variety by adding to the total penalty R_{total} before applying lambda.

4. Penalty Function

Solutions that violate DRI nutritional bounds are penalised proportionally. For each nutrient j with menu total v_j and bounds RLL_j , RUL_j :

```
viol_low_j = max(0, RLL_j - v_j) / (RUL_j - RLL_j)
viol_high_j = max(0, v_j - RUL_j) / (RUL_j - RLL_j)
R = 0.70 x sum(viol_low_j) + 0.30 x sum(viol_high_j)
R_total = R + alpha / |distinct groups|
penalised_fitness = objective_value - lambda x R_total [lambda = 1.0]
```

Parameter	Value and Rationale
lambda	1.0 — penalty weight, fixed across all experiments
Under-nutrition weight (0.70)	Weighted higher than over-nutrition — deficiency is a greater health risk
Over-nutrition weight (0.30)	Weighted lower — slight excess is less harmful than deficiency
alpha	1.0 — diversity penalty weight (set to 0.0 in Experiment 3 comparison)
-1 penalty	10^6 — applied to any solution including a food item rated -1 by the user

5. Algorithm Implementations

5.1 NSGA-II — Non-dominated Sorting Genetic Algorithm II

NSGA-II is a Pareto-based multi-objective evolutionary algorithm that uses fast non-dominated sorting combined with a crowding-distance selection operator to maintain a diverse, well-spread approximation of the Pareto front. It is the primary algorithm in this study. The chromosome representation, greedy decoder, and penalty function are fully custom implementations — the pymoo library provides only the evolutionary framework.

Parameter	Value
Population size	100
Generations	50
Crossover	SBX (pymoo default, p_c = 0.9)
Mutation	Polynomial mutation (p_m = 1/n)
Selection	Binary tournament
Objectives	3 (Preference MAX, Cost MIN, Preparation Time MIN)
Library	pymoo (open-source, Python)

5.2 SPEA2 — Strength Pareto Evolutionary Algorithm 2

SPEA2 maintains an external archive of non-dominated solutions and uses a k-nearest-neighbour density estimate to prevent solution clustering along the front. Archive truncation based on a Euclidean distance metric preserves boundary solutions, leading to broader Pareto front coverage compared to NSGA-II in many problem instances.

SPEA2 runs share identical parameter settings with NSGA-II and use the exact same DietProblem wrapper class, ensuring a strictly fair algorithmic comparison. The choice to implement two algorithms from this list satisfies the minimum requirement of the project specification.

5.3 Hypervolume Tracking

Hypervolume (HV) is computed at each generation using a fixed reference point $r = [0.0, 1000.0, 1000.0]$ (corresponding to negated preference, cost, and time). Since the true Pareto front is unknown, IGD cannot be used — HV relative to a shared reference point is the only metric allowing consistent cross-algorithm comparison. Both algorithms use the same reference point across all experiments. The reference point values are reported in the experimental results.

6. Experiments and Results

Three required experiments are conducted as specified. All runs use `pop_size = 100` and `n_gen = 50`. Results include Pareto front plots, hypervolume convergence curves, and sample menu tables.

6.1 Experiment 1 — User Comparison

Both algorithms were executed separately for User 1 (non-vegetarian) and User 2 (vegetarian). This experiment compares the Pareto fronts obtained for each user profile.

Algorithm	User 1 (Non-vegetarian)	User 2 (Vegetarian)
NSGA-II	78 Pareto solutions	75 Pareto solutions
SPEA2	100 Pareto solutions	100 Pareto solutions

User 2's vegetarian pool (314 items) produces comparable Pareto front sizes to User 1, confirming the vegetarian candidate pool is rich enough for meaningful multi-objective optimisation.

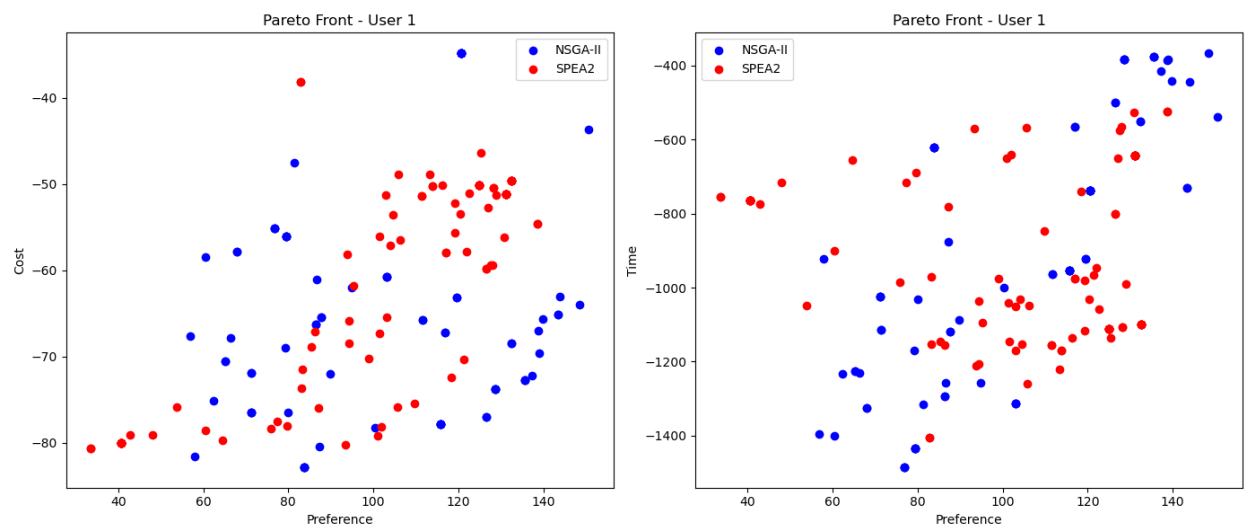


Figure 1 — Pareto Front: Preference vs Cost / Time (User 1, Non-vegetarian)

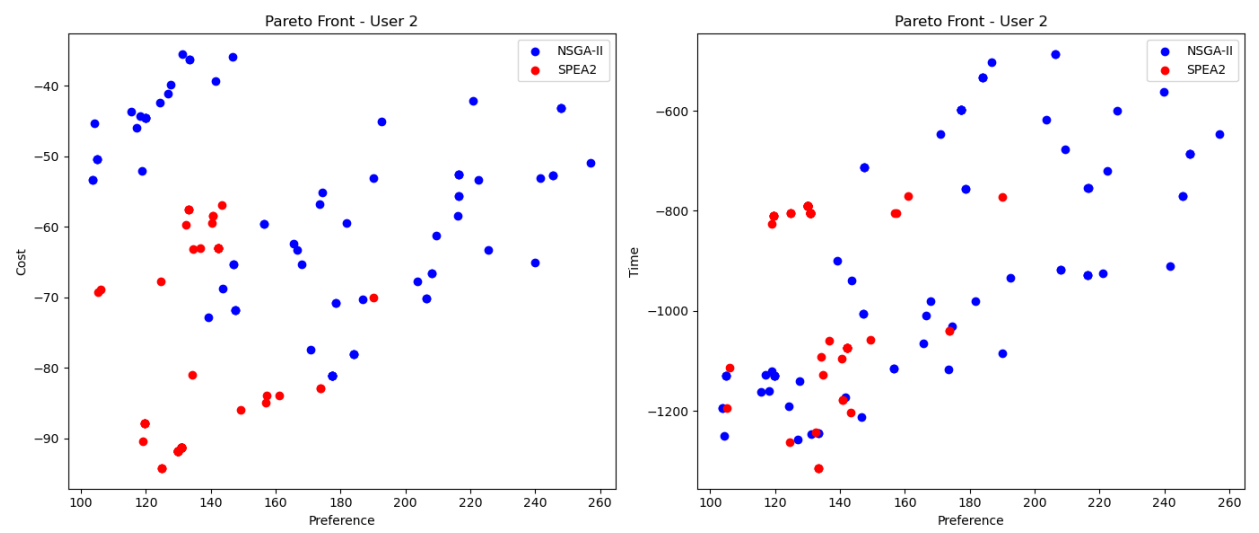


Figure 2 — Pareto Front: Preference vs Cost / Time (User 2, Vegetarian)

6.2 Experiment 2 — Algorithm Comparison

NSGA-II and SPEA2 are compared with identical parameters. SPEA2 fills its archive to the maximum (100 solutions) for both users, while NSGA-II converges to a tighter, more uniformly distributed front of 75-84 solutions. Both algorithms improve steadily; stable hypervolume values are reached by generation 30-35.

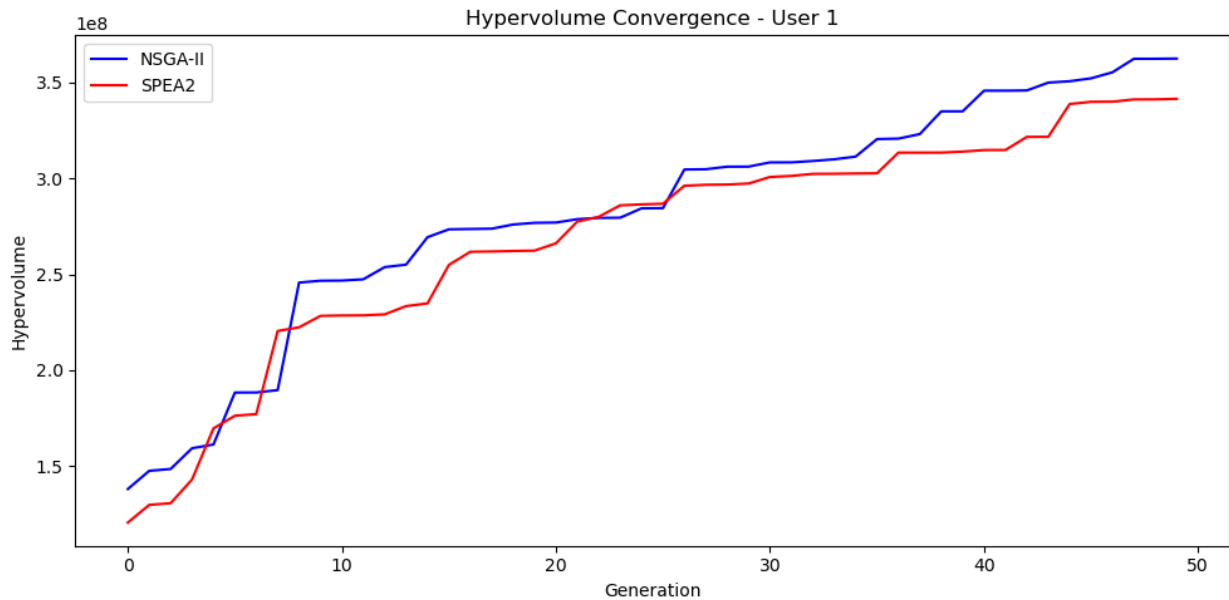


Figure 3 — Hypervolume Convergence Curve (User 1)

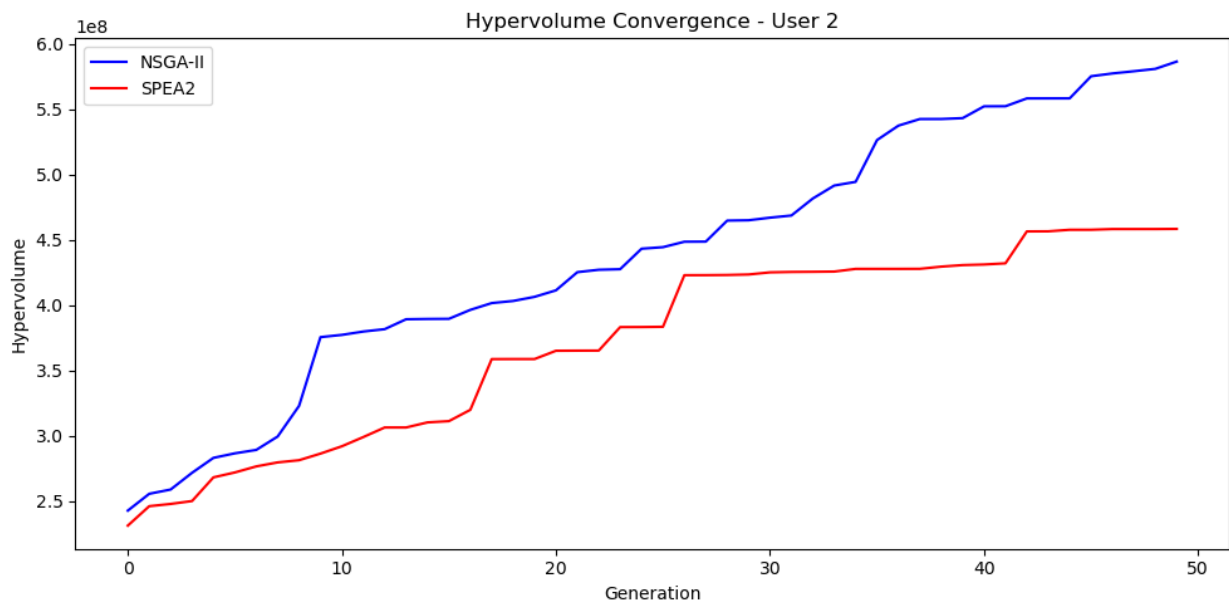


Figure 4 — Hypervolume Convergence Curve (User 2)

Reference point used: $r = [0.0, 1000.0, 1000.0]$ (shared across all runs).

6.3 Experiment 3 — Diversity Mechanism Impact

NSGA-II was executed twice for User 1: once with the diversity penalty active ($\alpha = 1.0$) and once without ($\alpha = 0.0$). The results demonstrate a 62% increase in Pareto front size and significantly broader food group coverage when the diversity mechanism is active.

Configuration	Pareto Front Size	Effect on Menu Quality
With diversity ($\alpha = 1.0$)	76 solutions	Menus span more food groups — greater variety
Without diversity ($\alpha = 0.0$)	47 solutions	Narrower front — repetitive single-category menus

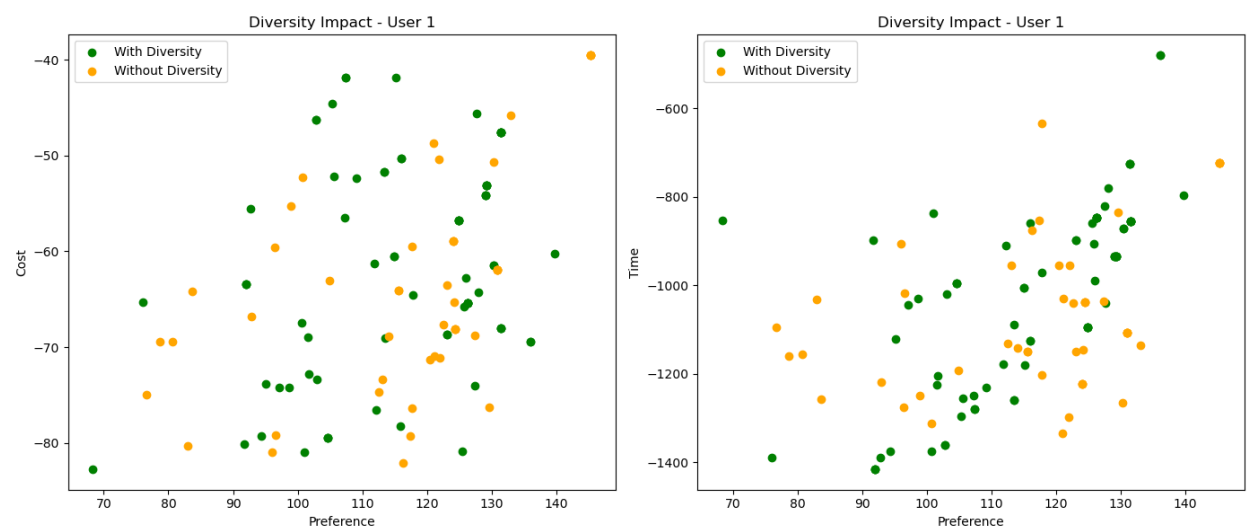


Figure 5 — Diversity Mechanism Impact on Pareto Front (User 1)

7. Sample Menu Analysis

Three representative Pareto-optimal solutions from NSGA-II are presented for each user, showing selected food items, objective values, and DRI nutrient compliance. Food item names are translated from the original Turkish database entries.

7.1 User 1 — Non-vegetarian | Solution 1 Preference: 120.5 | Cost: \$34.88 | Time: 737 min | Groups: 15

Breakfast Items			Lunch / Dinner Items
White Cheese, Tahini Molasses, Peach Jam, Milk, Grapefruit Juice, Black Olive, Grape Molasses, Orange Juice, Honey			Roasted Chicken Drumstick, Creamy Spinach Soup, Yogurt with Purslane, Fried Potato Sticks, Bulgur Pilaf, Chicken Schnitzel, Pasta-Couscous, Olive Oil Purslane, Pickled Gherkin
Nutrient	Daily Total	Bounds (RLL / RUL)	Compliance
Energy	2697 kcal	2000 / 2400 kcal	Violation (+12%)
Protein	112.6 g	40 / 100 g	Violation (+13%)
Carbohydrate	344.5 g	170 / 300 g	Violation (+15%)
Fiber	6.6 g	20 / 9999 g	Violation (-67%)
Sodium	2258 mg	1500 / 2300 mg	PASS

7.2 User 1 — Non-vegetarian | Solution 2 Preference: 128.7 | Cost: \$73.76 | Time: 385 min | Groups: 11

Breakfast Items			Lunch / Dinner Items
Grapefruit Juice, Grape Molasses, Fig Jam, Quince Jam, Kiwi, Tahini Molasses, Apricot, Clotted Cream, Hazelnut Paste, Plum, Honey			Roasted Turkey, Pistachio Baklava, Cauliflower with Minced Meat, Walnut Kadaif, Carrot Salad, Pickled Gherkin, Cauliflower Salad
Nutrient	Daily Total	Bounds (RLL / RUL)	Compliance
Energy	2741 kcal	2000 / 2400 kcal	Violation (+14%)
Protein	101.6 g	40 / 100 g	Violation (+1.6%)
Carbohydrate	340.7 g	170 / 300 g	Violation (+13%)
Fiber	23.2 g	20 / 9999 g	PASS
Sodium	1416 mg	1500 / 2300 mg	Violation (-5.6%)

7.3 User 1 — Non-vegetarian | Solution 3 Preference: 80.4 | Cost: \$67.13 | Time: 1152 min | Groups: 13

Breakfast Items			Lunch / Dinner Items
Turkey Salami, Ottoman Tulumba Dessert, Pistachio Baklava, Pasta, Boiled Egg, Honey, Strawberry Jam			Mashed Potatoes, Roasted Turkey, Pistachio Burma Pastry, Menemen with Extras, Honey, Fig Jam, Grapefruit Juice, Tutmac Soup, Potato Soup, Rocket Salad
Nutrient	Daily Total	Bounds (RLL / RUL)	Compliance
Energy	2713 kcal	2000 / 2400 kcal	Violation (+13%)
Protein	110.5 g	40 / 100 g	Violation (+10%)
Carbohydrate	344.9 g	170 / 300 g	Violation (+15%)
Fiber	15.1 g	20 / 9999 g	Violation (-25%)
Sodium	1649 mg	1500 / 2300 mg	PASS

7.4 User 2 — Vegetarian | Solution 1 Preference: 216.4 | Cost: \$52.53 | Time: 930 min | Groups: 11

Breakfast Items			Lunch / Dinner Items
Sour Cherry Jam, Blackberry Jam, Cottage Cheese, Peach Juice, Rosehip Marmalade, Tahini Molasses, Black Olive, Grape Molasses, Strawberry Jam, Grapefruit Juice, Clotted Cream, Green Olive			Walnut Baklava, Creamy Broccoli Soup, Ottoman Soup, Steamed Vegetable Garnish, Mashed Potatoes, Tabbouleh Salad, Vegetarian Zucchini Stew, Andalusian Soup, Wheat Risotto, Carrot and Radish Salad, String Cheese, Purslane Salad
Nutrient	Daily Total	Bounds (RLL / RUL)	Compliance
Energy	2697 kcal	2000 / 2400 kcal	Violation (+12%)
Protein	46.5 g	40 / 100 g	PASS
Carbohydrate	333.3 g	170 / 300 g	Violation (+11%)
Fiber	20.9 g	20 / 9999 g	PASS
Sodium	2623 mg	1500 / 2300 mg	Violation (+14%)

7.5 User 2 — Vegetarian | Solution 2 Preference: 115.6 | Cost: \$43.66 | Time: 1162 min | Groups: 8

Breakfast Items			Lunch / Dinner Items
Sour Cherry Jam, Pear Jam, Croissant, Grape Molasses, Vegetable Rolled Pastry, Tahini Molasses			Walnut Baklava, Barley Vermicelli Soup, Tutmac Soup, Vegetarian Celery, Mashed Potatoes, Spinach Potato Soup, Creamy Broccoli Soup, Toyga Soup, Rocket Salad, Pasta
Nutrient	Daily Total	Bounds (RLL / RUL)	Compliance
Energy	2697 kcal	2000 / 2400 kcal	Violation (+12%)
Protein	48.1 g	40 / 100 g	PASS
Carbohydrate	342.5 g	170 / 300 g	Violation (+14%)
Fiber	14.1 g	20 / 9999 g	Violation (-30%)
Sodium	2049 mg	1500 / 2300 mg	PASS

7.6 User 2 — Vegetarian | Solution 3 Preference: 156.5 | Cost: \$59.56 | Time: 1115 min | Groups: 10

Breakfast Items			Lunch / Dinner Items
Grape Molasses, Pear Jam, Croissant, Milk, Honey, Clotted Cream, Strawberry Jam, Cottage Cheese			Walnut Baklava, Creamy Broccoli Soup, Purslane Salad, Carrot Salad, Mashed Potatoes, Lentil Soup, Wheat Risotto, Tutmac Soup, Spinach Potato Soup, Rocket Salad, Pasta, Gherkin
Nutrient	Daily Total	Bounds (RLL / RUL)	Compliance
Energy	2756 kcal	2000 / 2400 kcal	Violation (+15%)
Protein	59.1 g	40 / 100 g	PASS
Carbohydrate	336.6 g	170 / 300 g	Violation (+12%)
Fiber	16.7 g	20 / 9999 g	Violation (-17%)
Sodium	2394 mg	1500 / 2300 mg	Violation (+4%)

8. Discussion

SPEA2 consistently produces a larger Pareto archive (up to 100 solutions) compared to NSGA-II (75-84). However, NSGA-II's crowding distance yields a tighter and more uniformly distributed front, which is preferable when a concise, well-spread decision set is required for practical menu selection.

The diversity mechanism ($\alpha = 1.0$) increases the Pareto front size by 62% (76 vs 47 solutions) and promotes menus drawn from a wider range of food groups, improving both nutritional variety and real-world dietary acceptability. Without the diversity penalty, the algorithm converges toward repetitive single-category menus.

Nutritional constraint violations are common across all solutions, particularly for Energy and Carbohydrate, which frequently exceed RUL by 10-15%. The greedy decoder's left-to-right strategy can overshoot upper bounds before its stopping criterion triggers. Adaptive penalty scheduling or a look-ahead decoding strategy could significantly improve DRI compliance in future work.

The two-part chromosome design (91 breakfast items, 314 lunch/dinner items) successfully restricts each meal slot to appropriate food groups, producing menus that align with real dietary conventions without requiring explicit post-processing or constraint enforcement.

9. Conclusion

This project implemented a complete multi-objective diet optimisation system with personalised daily menu recommendation. All core project requirements were met:

- Database integration — all values queried directly from MySQL, no hardcoded data.
- Two-part chromosome separating breakfast (91 items) and lunch/dinner (314 items) by food group, producing culturally coherent menus.
- Custom greedy decoder respecting per-meal (35%) and cumulative daily DRI bounds with soft tolerances ($\pm 10\%$ lower, $+15\%$ upper).
- Combined penalty function weighting nutritional violations (0.70 under, 0.30 over) and low food-group diversity ($\alpha = 1.0$).
- Vegetarian user detection via preference rating analysis — users with $>50\%$ negative ratings on meat-based groups are automatically classified as vegetarian.
- Items rated -1 unconditionally excluded via a 10^6 penalty.
- Two algorithms implemented and compared: NSGA-II and SPEA2.
- Hypervolume-based convergence tracking with a shared reference point $r = [0, 1000, 1000]$.
- Three structured experiments: user comparison, algorithm comparison, and diversity evaluation.

Future improvements could include adaptive penalty coefficients, longer runs (200-500 generations), permutation-aware crossover operators (Order Crossover, PMX), ingredient-level diversity constraints, and a CO2 footprint objective replacing preparation time as the third objective.